

Group E: Final Materials Report

Carlotta Bogena,¹ Oliver Hammond,¹ Ira Hanabusa,¹ William Harris,¹ Laura Jian,¹ and Ziguang Zhu¹
University of California, San Diego

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The goal of this project is to research appropriate materials required to build the Accessory Dwelling Units. This research will be used by Triton Solar Decathlon in the Orange County Sustainability Decathlon competition. The ADU will be built in San Diego for the residents located in Logan Heights. Materials researched in this report must meet a certain set of criterion. The housing components that this report will focus on are: flooring, roofing, and the interior and exterior walls.

I. INTRODUCTION

Triton Solar Decathlon requires an in-depth analysis of affordable, sustainable, and human-safe materials to reduce environmental and financial costs of constructing homes. Triton Solar Decathlon needs materials that will help in their competition to create affordable, environmentally-friendly, and safe ADUs for the residents at Logan Heights.

Our group has researched a set of materials for each housing component, which include: roof, floor, and interior and exterior wall. The materials for each component are compared and rated using a set of criterion that was decided. The criterion includes: affordability, desirability, producibility, ecological sustainability, usability, and architectural fit. In addition, there is an associated spreadsheet of costs, explanation of ratings and a weighted decision matrix. These assessments can be located in our [folder](#).

A. Residents at Logan Heights

The affected community are the residents of Logan Heights, they are usually multigenerational low-income families. Eighty-eight percent of the residents at Logan Heights are Hispanic. The average family size at Logan Heights is four and the median household income is \$43.858 thousand per year. Compared to this, the median household income in San Diego is \$63.739 thousand per year. Logan Heights, along with the rest of the San Diego area, suffers from the lack of affordable housing. Among many other factors, this is largely due to gentrification, as Logan Heights the demographics of the neighborhood have changed to include wealthier people from outside areas, increasing the value of real estate, and the price of housing with it. Real estate prices have increased from about \$250 thousand to \$565 thousand since 2012. Furthermore, this has not only displaced lower-income individuals but increased the homeless population. This problem stems from redlining, which is the practice of banks, insurance companies, and other finance institutions refusing or limiting loans, mortgages, insurance payouts, etc, within specific geographic areas, particularly low income neighborhoods, which often have a disproportionate number of racial minority groups. San Diego, and specifically, Logan Heights is one such area which has been redlined. Community action and organizations can help solve this problem. Nonprofits need to start constructing the much needed low-income hous-

ing while local voters need to fight to pass legislation to prohibit luxury project development in at-risk negative neighborhoods and increase funding for home-repair programs focusing on gentrified neighborhoods. This will help decrease the effects of redlining and gentrification in San Diego.

II. CRITERION

In order to compare each material a set of criterion was decided. The criteria chosen were: affordability, usability, producibility, ecological sustainability, desirability, and architectural fit (listed in order of importance).

Criterion	Requirement
Affordability	For Triton Solar Decathlon, the final design should stay under the budget of \$180,000 for building materials of an ADU.
Usability	Structurally sound and requires low maintenance. Materials should not emit toxic fumes and particles harmful to residents.
Producibility	Chosen materials should be in abundance and can be sourced from local manufacturers or providers.
Ecological Sustainability	Opt for materials that last long or materials that can be repurposed or recycled.
Desirability	ADUs should be visually inviting. Consider the exterior and interior aesthetics.
Architectural Fit	Structure fits into the local environment and is well suited for the climate of the area.

FIG. 1. Table of criteria with explanation.

Affordability was chosen to be our most important criteria because budget is one of our main constraints: no matter how many other criteria a material fits, if it does not fit into the budget it can not be purchased and used. Usability is second, it takes into account the daily interactions the residents have with the materials, this criteria takes into consideration the potential health hazards each material could have on the residents. Producibility and ecological sustainability are both in third. They are similar and we wanted to have materials that could be sourced locally (reducing carbon footprints) and friendly to the environment. In addition, we want it so that if this were mass produced, materials would be easily accessible in the community. Lastly we have architectural fit. Most materials that do not pose health hazards and are environmentally friendly, are already fitting for the state of California, these factors are already included in the other criteria. But in addition, this criteria takes into account the local climate of the area, in this case Logan Heights when rating the materials.

III. ASSESSMENTS

The three main ways that were used to assess the materials for each housing component are the spreadsheet of costs, explanation of ratings, and the weighted decision matrix. These assessments could be found in our [assessment folder](#).

A. Spreadsheet of Costs

The spreadsheet of costs is the associated cost for each material. This contains the cost per square feet, the installation costs, and the total costs for using the material in the competition. For this, a lower and upper limit of costs were explored and averaged towards the end. The spreadsheet of costs were used in rating the materials for the affordability criteria.

B. Weighted Decision Matrix

The weighted decision matrix is a table of ratings for each material and for each criteria. It is a way to evaluate all materials within a housing component and discern which material best covers all of the criteria.

In the decision matrix, the more important the criteria, the more heavily it will be weighted. Each criteria for each material is rated out of five. The final score for each material is simply the sum of the individual weighted scores, where the individual weighted scores are the weighting multiplied by the weight for that criteria. The material with the "best" or highest score is the material that best meets all of our listed criteria.

Housing Component Materials					
Criteria	Weight	Option One		Option Two	
		Rating (out of 5)	Total	Rating (out of 5)	Total
Affordability	30	5	150	2	60
Desirability	10	5	50	1	10
Producibility	15	5	75	2	30
Ecological Sustainability	15	2	30	1	15
Usability	25	3	75	1	25
Culturally Appropriate	5	3	15	3	15
		SCORE	395	SCORE	155

FIG. 2. Example weighted decision matrix where option one is the best material.

1. Uses

The decision matrix is best used as a guideline to choose which material is the most suitable based on the criteria listed above. The material with the best overall score would indicate that that material best covers the set of criterion. In using the decision matrix this way, only the overall score is taken into account and the individual ratings for each criteria is not. However, the decision matrix could also be used to only look at a specific criteria. In this case, the material with the best rating for that criteria would be most suitable.

The decision matrix acts only as a guide, the ultimate decision will be made by the Triton Solar Decathlon Team. Weights and ratings on the decision matrices can be changed as needed by Triton Solar Decathlon to suit different goals.

C. Explanation of Ratings

The explanation of ratings is the explanation of the ratings for each material for each criteria.

IV. HOUSING COMPONENT

For each housing component, a weighted decision matrix was created to compare each material, the overall scores of all materials are highlighted in yellow and the material with the highest score is highlighted in red.

A. Roof

Criteria	Weight	Metal Roofing		Asphalt Composite Shingles		Slate		Clay Tile		Wood Shingles		TPO	
		Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total
Affordability	30	5	150	4	120	1	30	4	120	4	120	5	150
Desirability	10	2	20	3	30	5	50	3	30	2	20	1	10
Producibility	15	4	60	4	60	2	30	4	60	4	60	3	45
Ecological Sustainability	15	4	60	1	15	2	30	3	45	4	60	2	30
Usability	25	5	125	5	125	4	100	3	75	2	50	5	125
Architectural Fit	5	5	25	5	25	5	25	5	25	2	10	2	10
		SCORE	440	SCORE	375	SCORE	265	SCORE	365	SCORE	320	SCORE	370

FIG. 3. Final weighted decision matrix for primary roofing material

For the primary roofing material, we considered a variety of options. The materials we chose to compare for the final product were metal, slate, clay tile, asphalt shingles, wood shingles, and TPO. Due to the temperate Mediterranean climate of San Diego, weather was not a huge deciding factor in choosing the best material unlike if the house were constructed in a more hostile environment. Of the materials, our best options came down to metal, asphalt shingles, and clay tile. These three are extremely common across Southern California and would allow the unit to fit in, regardless of the choice. All have decent insulation values and hold up well if extreme weather were to impact them. All are also relatively cheap in comparison to other options. Ultimately, the team chose to use metal for the final project as it was deemed the most applicable in this scenario.

1. Summary of Ratings

Judging based entirely on the affordability of materials, metal roofing ranks only third with asphalt shingles and TPO both being cheaper options. However, it is still relatively cheap, especially when using the least expensive option. It is also easy to obtain and would not need to be transported over long distances bringing transportation costs down. Because of this, metal roofing earns a 4 out of 5. When it comes

Roof area:	46' - 8 5/16" long		23' - 4 5/16" wide		total		1091.32		sq ft											
material	cost / sq ft low	cost / sq ft high	cost - low	cost - high	installation / sq ft low	installation / sq ft high	total install low	total install high	TOTAL - LOW (best case scenario)	TOTAL - HIGH (worst case scenario)	Average									
Metal	\$150	\$25	\$1,636.98	\$27,283	\$4.50	\$9	\$4,910.94	\$5,822	\$6,547.92	\$37,105	\$21,826.40									
Slate	\$10	\$22	\$10,913	\$24,009	\$6	\$10	\$6,548	\$10,913	\$17,461	\$34,922	\$26,192									
Clay Tile	\$3	\$8	\$3,274	\$8,731	\$4	\$16	\$4,365	\$17,461	\$7,639	\$26,192	\$16,915									
Asphalt Shingles	\$2	\$4	\$2,183	\$4,365	\$2	\$3.50	\$2,183	\$3,819.62	\$4,365	\$8,185	\$6,275									
Wood Shingles	\$3	\$5.50	\$3,274	\$6,002.26	\$5	\$10	\$5,457	\$10,913	\$8,731	\$16,915.46	\$12,823									
TPO	\$2	\$3.50	\$2,183	\$3,819.62	\$3.50	\$5.25	\$3,819.62	\$5,729.43	\$6,002	\$9,549.05	\$7,776									

FIG. 4. Cost assessment for primary roofing material

to desirability, there are significantly more desirable options available. Even the more expensive versions of metal roofing are better looking than the option the team decided on. The only problem is that desirability and aesthetics are generally proportional to cost. Due to these determinations, we gave it a 2 out of 5 for the desirability rating. Considering how producible metal is, it ranks highest among the materials we decided from. While it requires a decent amount of energy to produce, its common use in the area means it is produced locally. Also, the materials needed for production are easily obtained and can be made from other recycled products so we decided to give it a 4 out of 5. One of the more important factors in the consideration of materials for the decathlon is the ecological sustainability of the options. As previously stated, metal can be recycled and does not require a large amount of energy which results in a lower environmental impact. Also, metal does not generate any toxins that could be harmful to the residents of the unit and those around them. Based on these realizations, we gave it a 4 out of 5 as it is not quite perfect due to the initial extraction and purification of metal not being quite perfect environmentally. When it comes to usability, metal gets a perfect score. It has a long lifespan and can be easily replaced. It also holds up well in weather and has decent insulation. Another realization is that metal is very easy to put solar panels on top of, which is a primary component of the requirements of the unit. Our final criteria and least impactful in our decision is the architectural fit of the material and of course it achieves a perfect score. Many units in the San Diego area utilize metal roofing so the unit will fit into its surroundings seamlessly.

B. Floor

Criteria	Weight	Weighted Decision Matrix: Flooring											
		Polished Concrete			Linoleum			Bamboo			Cork		
Affordability	30	5	150	5	150	3	90	3	90	3	90	1	30
Durability	10	5	50	3	30	5	50	3	30	3	30	5	50
Productivity	15	5	75	2	30	4	60	2	30	2	30	5	75
Ecological Sustainability	15	2	30	5	75	4	60	5	75	5	75	5	75
Usability	25	5	125	3	75	4	100	4	100	2	50	3	75
Architectural Fit	5	5	25	5	25	5	25	3	15	3	15	5	25
		SCORE	450	SCORE	350	SCORE	350	SCORE	340	SCORE	290	SCORE	310

FIG. 5. Final weighted decision matrix for primary flooring material

For the flooring, linoleum, ceramic tile, bamboo, cork, hardwood, and polished concrete were compared. Linoleum and hardwood were both original potential flooring materials chosen by the Triton Solar decathlon team. Bamboo and cork were selected as potentially more sustainable alternatives for hardwood. Bamboo can be harvested at 4-7 years of age while hardwood trees must grow for at least 20 years. Cork is harvested from the bark of the cork oak tree without damaging

the tree itself, which can be harvested again after 9 years of bark growth. And finally, ceramic tile and polished concrete were selected due to their durability and the fact that they are easy to produce and created out of naturally abundant materials.

1. Summary of Ratings

Floor area:	46' - 8 5/8" long 23' - 4 5/16" wide total				1091.32 sq ft														
material	cost / sq ft low	cost / sq ft high	cost - low	cost - high	installation / sq ft low	installation / sq ft high	total install low	total install high	TOTAL - LOW (best case scenario)	TOTAL - HIGH (worst case scenario)	Average								
Linoleum	\$4.00	\$5	\$4,362.28	\$5,751	\$0.41	\$1	\$447.44	\$768	\$4,812.72	\$6,519	\$5,674.96								
Ceramic Tile	\$3	\$6	\$3,274	\$6,548	\$13	\$23	\$14,107	\$25,100	\$17,461	\$31,948	\$24,555								
Bamboo	\$5	\$8	\$5,457	\$8,731	\$6	\$10	\$6,548	\$10,913	\$12,005	\$19,644	\$15,824								
Cork	\$5	\$10	\$5,457	\$10,913	\$2	\$3.00	\$2,183	\$3,273.96	\$7,639	\$14,187	\$10,913								
Hardwood	\$2	\$6.00	\$2,183	\$6,547.92	\$6	\$12	\$6,548	\$13,369	\$8,731	\$19,916.59	\$14,324								
Polished Concrete	\$5	\$5.72	\$5,129	\$6,242.35	\$2.00	\$2.20	\$2,182.64	\$2,400.80	\$7,312	\$8,643.25	\$7,978								

FIG. 6. Cost assessment for primary flooring material

The most affordable materials out the six are by far linoleum and polished concrete, with average costs of \$5.5 and \$8 thousand respectively. The average costs of cork, hardwood, and bamboo were relatively similar and ranged from \$11-\$16 thousand. And by far the most expensive option is ceramic tile at \$25 thousand due to high labor costs.

In terms of desirability, hardwood wins by a long shot because its culturally seen as it is aesthetically pleasing and is considered luxurious. All of the other materials have some reasons to be undesirable. Linoleum is considered old-fashioned, polished concrete and ceramic tile can be cold, and bamboo and cork do not have many design options.¹

Polished concrete and ceramic tile are by far the most producible materials as they are created using naturally abundant materials and can be locally produced. Hardwood is almost as producible, however, one must look for local hardwood species and ensure the product is FSC certified to guarantee it was harvested sustainably. Meanwhile, cork, bamboo, and linoleum are all sustainably producible, however, none are locally produced.²

For Environmental Sustainability, Bamboo, cork, and ceramic tile all tied for the highest score in this category. Bamboo and cork floors will last about 50 years and are both recyclable materials. Furthermore, as mentioned previously mentioned, both are highly sustainable since they can be harvested much faster than wood and harvesting cork doesn't harm the tree it comes from. Ceramic tile lasts 30 years and uses low energy in its production, and produces no carbon emissions. Hardwood ranked just under these three because it lasts 50 years and can be often repurposed and recycled, however, one must source local species for it to be sustainable. And finally, polished concrete ranked very low despite its 100+ year life span due to its high carbon emissions.³

As for usability, polished concrete ranked the highest because it is strong, durable, and easy to maintain. Next Bamboo and hardwood are very durable, but prone to water and termite damage and require more maintenance than polished concrete. Cork and linoleum also suffer from being damage prone from water, light, and sharp or heavy objects. And finally, ceramic tile would rank as high as polished concrete for

the same reasons of being durable and easy to maintain if it weren't for the fact that they often contain lead particles.⁴

Polished concrete, linoleum, hardwood, and ceramic tile all received the same high score for cultural appropriateness since they are all commonly used materials in the US, while cork and bamboo received lower scores due to not being as commonly used in the US.

After being scored appropriately in all above categories, polished concrete scored the highest, followed by linoleum, ceramic tile, hardwood, bamboo, and cork.

C. Interior Walls

Criteria	Weight	Interior Wall							
		Gypsum Plaster		Venetian Plaster		Gypsum Dry Walls		Wooden Wall Panels	
		Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total	Rating (out of 5)	Total
Affordability	30	3	90	2	60	5	150	3	90
Desirability	10	4	40	5	50	4	40	5	50
Producibility	15	5	75	5	75	5	75	3	45
Ecological Sustainability	15	5	75	4	60	5	75	5	75
Usability	25	2	50	5	125	2	50	3	75
Architectural Fit	5	5	25	5	25	5	25	3	15
		SCORE	355	SCORE	395	SCORE	415	SCORE	350

FIG. 7. Final weighted decision matrix for interior walls

For the interior walls, gypsum dry wall, gypsum plaster, Venetian plaster, and wooden wall panels were chosen as the materials for comparison. Gypsum plaster is an original potential interior wall material that Triton Solar Decathlon has chosen. Since wall plasters are a common interior wall material, Venetian plaster was chosen to be compared to gypsum plaster, while it is more expensive for plasters, it also holds many benefits that other materials do not. Dry walls are another common interior wall material, so gypsum dry wall was chosen to be compared. It was also chosen because it contains the same material as the gypsum plasters that was an original potential material. Lastly, wooden wall panels was chosen, while it is not a common interior wall material due to the price and the cost of upkeep, it was chosen due to being

Venetian plaster rates the best in usability and contains the most benefits. It comes in second, after Gypsum Dry Wall with its lowest rating in the affordability category. Venetian plaster is made of lime which is "naturally high in PH, so it automatically acts as an antibacterial surface, which inhibits the growth of mold and other fungus.⁵" It is also flame resistant, low maintenance, breathable and mildew-resistant. And emits zero VOCs, where exposure can cause health effects, like irritation, headaches, nausea, etc.

Gypsum Plaster comes in third. Similar to the dry wall, gypsum dust can irritate parts of the body. In addition gypsum is a toxin in landfills. However, gypsum is abundant and deposits can be found in many parts of the world. It is natural, light-weight mineral deposited across the major sedimentary rocks. It is highly durable material and has a great ability to withstand fire, corrosion, and heat.

Lastly, the wooden wall panels scores the lowest in the decision matrix. It is the most expensive out of all materials and over ten times the cost of gypsum dry wall. It runs the risk of fire, but is accessible: it could be obtained and bought from local home good stores, however certain types of wood could be harder to obtain. But if sourced properly they can be very ecologically friendly since wood is natural, recyclable and biodegradable.

D. Exterior Walls

Weighted Decision Matrix: Exterior Wall					
Criteria	Weight	Fiber-cement		Insulated Vinyl	
		Rating (out of 5)	Total	Rating (out of 5)	Total
Affordability	30	4	120	4	120
Desirability	10	5	50	4	40
Producibility	15	5	75	5	75
Ecological Sustainability	15	4	60	4	60
Usability	25	4	100	3	75
Architectural Fit	5	5	25	5	25
		SCORE	430	SCORE	395

FIG. 9. Final weighted decision matrix for exterior siding

1. Summary of Ratings

Outer Interior Walls	48' x 10' (2)		23' x 10' (2)		Total Square Feet		(190)					
material	cost / sq ft low	cost / sq ft high	cost - low	cost - high	installation / sq ft low	installation / sq ft high	total install low	total install high	TOTAL - LOW (best case scenario)	TOTAL - HIGH (worst case scenario)	Average	
Gypsum Plaster	\$10.00	\$11.00	\$13,800.00	\$15,180.00	\$6.14	\$6.21	\$7,093.20	\$8,560.80	\$20,893.20	\$23,749.80	\$22,321.50	
Venetian Plaster	\$10.00	\$25.00	\$13,800.00	\$34,500.00	\$6.31	\$7.52	\$7,327.80	\$10,377.60	\$21,127.80	\$44,877.60	\$33,002.70	
Gypsum Dry Wall	\$0.45	\$0.65	\$621.00	\$807.00	\$1.50	\$3.00	\$2,070.00	\$4,140.00	\$2,691.00	\$5,037.00	\$3,864.00	
Wood Wall Panels	\$12.00	\$40.00	\$18,000.00	\$55,200.00	\$6.80	\$9.86	\$9,384.00	\$13,608.80	\$25,944.00	\$58,809.80	\$47,376.40	

FIG. 8. Cost assessment for interior siding.

In terms of affordability gypsum dry wall is the obvious choice, with an average cost of around \$4 thousand, while all other materials are over five times as expensive.

Gypsum Dry Wall comes in first in overall score. Besides its cheap cost, it is 100% recyclable, has a small carbon footprint, and is light and can be reused. It is also fire resistant and flexible. However, a caution in dry walls is that breathing the dust from join compounds may cause throat and airway irritation, which could lead to breathing difficulties. While this is an issue, this mostly applies to installation and renovation.

Initial considerations for exterior siding included fiber-cement, insulated vinyl, stucco, and ipe wood. Fiber-cement is a durable, long-lasting siding option composed of Portland cement, sand, and cellulose. Vinyl is a common siding option across the United States and is made from polyvinyl chloride, the same thermoplastic used in PVC pipes. The added polystyrene layer beneath the plastic dramatically increases insulation and impact resistance for a cheap price. Stucco—composed of cement, sand, and water—is similar to fiber-cement in terms of composition and strength. However, it was eventually discounted due to its porous texture that allows moisture absorption which causes molding and discoloration. Since Logan Heights neighbors the coast, the area has a humid climate that makes stucco unsuitable as a siding option. Wood also suffers from its vulnerability to water damage. Ipe, or Brazilian walnut, is an exception because of its density and natural resistance to rot, abrasion, and weather. However, the luxurious quality of ipe makes it difficult and expensive to source, an unfortunate consequence considering

that affordability is the largest priority of the project. Therefore the final comparison was set between fiber-cement and insulated vinyl.

1. Summary of Ratings

Exterior Wall					Total Square Feet	1401.56					
material	material cost / sq ft low	material cost / sq ft high	material cost low	material cost high	labor cost / sq ft low	labor cost / sq ft high	total labor low	total labor high	TOTAL LOW (best case)	TOTAL HIGH (worst case)	Average
fiber-cement	\$5.00	\$14.00	\$7,007.80	\$19,621.84	\$3.00	\$6.00	\$4,204.68	\$8,409.36	\$11,212.48	\$28,031.20	\$19,621
insulated vinyl	\$4.00	\$12.00	\$5,606.24	\$16,818.72	\$2.50	\$10.75	\$3,503.90	\$15,066.77	\$9,110.14	\$31,885.49	\$20,497
stucco	\$5.00	\$9.58	\$7,007.80	\$13,426.94	\$2.00	\$3.00	\$2,803.12	\$4,204.68	\$9,810.92	\$17,631.62	\$13,721
ipe wood	\$5.00	\$20.00	\$7,007.80	\$28,031.20	\$20.00	\$45.00	\$28,031.20	\$63,070.20	\$35,039.00	\$91,101.40	\$63,070

FIG. 10. Cost assessment for exterior siding

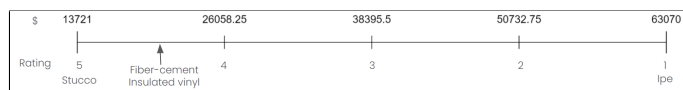


FIG. 11. Cost plot for exterior siding

Affordability was scored after assessing estimated costs for all considered siding options. Stucco is designated as the benchmark score of 5 for being the cheapest at \$13,700. On the other end, ipe is designated as the benchmark score of 1 for being the most expensive at \$63,070. Following stucco, fiber-cement and insulated vinyl both receive a 4 for being nearly equal in cost at \$20,000.

In terms of desirability, or visual aesthetic, fiber-cement has a slight edge over insulated vinyl. While fiber-cement is able to replicate the beauty of authentic wood, vinyl retains a cheap plastic texture no matter the application. This difference aside, both materials are available in unique siding styles such as shingle, lap, and clapboard and come in a variety of colors. Moreover, vinyl is a popular siding option nationwide so there is apparent merit to its sleek appearance.

Both fiber-cement and insulated vinyl receive top marks for producibility. Portland cement, sand, cellulose, polyvinyl chloride, and polystyrene are all abundant materials and can be sourced easily. In addition, there are a handful of services in the San Diego area that perform fiber-cement siding and insulated vinyl siding, so contracting work exterior siding will not be much of an issue.

For ecological sustainability, fiber-cement and insulated vinyl have their own share of pros and cons. Fiber-cement typically lasts up to 50 to 60 years. Portland cement, sand, and cellulose are all inert materials and are non-toxic for the environment. However, the amount of carbon emissions necessary to produce fiber-cement is greater than that of vinyl, and there are few recycling programs available.⁶ Insulated vinyl has a shorter lifespan of 20 to 40 years. Producing vinyl releases a non-degradable, highly toxic environmental pollutant known as dioxin, but the amount of dioxin emitted is minuscule compared to other contributors.⁷ Vinyl plastic is also completely recyclable. Overall, these materials are both green siding options.

Fiber-cement once again has an edge over insulated vinyl in the criteria of usability. Both are low maintenance options that generally require upkeep only to maintain appearance. The foam layer of insulated vinyl minimizes the thermal bridging and increases the impact resistance of standard vinyl siding, but its durability is still lower than fiber-cement. In addition, vinyl siding is prone to chipping and cracking and may require replacements within its life cycle. Thermoplastics are also known to outgas volatile organic compounds, but this is only a major concern for vinyl flooring which contaminates indoor air quality for a few weeks after installation.⁶

Finally, fiber-cement and insulated vinyl are both fitting options for the climate and surrounding architecture of the Logan Heights area. Fiber-cement is flame retardant with a class A fire rating, resistant to moisture and mold, pest resistant, weather resistant, and has good impact resistance. Insulated vinyl shares similar properties except with less impact resistance and a high insulation value that will lessen energy usage in the cold climate of the coast. Although vinyl is toxic when burning, fires are not a concern in the area.⁸ Around the site of construction, many of the homes employ a low profile color scheme of yellow, gray, and brown and have a wood or plaster-like texture for exterior siding. Fiber-cement and insulated vinyl are available in such colors and share a similar texture that would blend in well.

Overall, fiber-cement and insulated vinyl are both viable options for exterior siding, but when considering desirability and usability for the ADUs, fiber-cement comes ahead.

E. Insulation

Additional research was conducted for different insulation materials. This should serve as additional support and source of information for the TSD team. Special emphasis was placed on sustainable materials for the insulation of houses.

For insulation materials, the R-value plays a special role, as this indicates how well a material resists the conduction of heat; the higher the better at preventing heat conduction. One possible option is Stone mineral wool (Roxul) used for insulation. It uses recycled stone dust, comprising 75% to 90% recycled content. The main advantages are that it slows down spread of fire as it withstands temperature up to 1800 degrees, it is highly durable, versatile, and not affected by moisture, making it ideal for damp basements and humid climates. Additionally it has a R-value of 4 per inch, hence it meets the recommended insulation levels in California (Zone 3). Three to four inches of rock wool would be required in walls and fifteen inches in the attic. Roxul is available in batts and rigid panels with thicknesses from 1.25 inches. Comparing it to other materials it has a higher R-value than fiberglass and is easier to install and it also has great soundproofing. Another option is Hemp insulation. It is a renewable and natural friendly option. Hemp batting and hemp boards are used for insulation. One of the main advantages are that it is easy to recycle and reuse and requires little energy or water input to grow and manufacture. Its R-value is 2.5 per inch. Additionally it interacts well with moisture and is naturally hypoallergenic and non-toxic,

enhancing indoor air quality. Its main disadvantage is its lack of availability causing it to be more expensive. Additionally as it is resistant to compression and hard to cut its transportation is also expensive.

Unlike Hemp insulation Cellulose insulation can be produced locally, resulting in less emission due to transportation. It is also non-toxic and uses mostly recycled (80-100%) insulation material. Cellulose insulation consists of shredded newspaper and boric acid (safe, non-toxic but can cause eye and respiratory irritation) making it fire and insect resistant and antifungal. As it has very little air infiltration it serves as a great option for packing walls. Cellulose insulation is a very good ecofriendly insulation material due to its high R-value of 3.66 per inch, low environmental impact and high performance. Due to the dense packed cellulose it has a greater R-value than fiberglass and is more reliable than foam. However as it is sensitive to moisture it isn't suitable for damp environments.

Another option is Fiberglass insulation, which has a R-value of 2.2 to 3.8 per inch. It contains 20% recycled material, making it more eco-friendly than foam insulation. Additionally it is also more fire resistant than foam. Its main disadvantage is that it is difficult to install, can promote air infiltration and heat loss, and is sensitive to moisture.

A more easy to install option is Cotton insulation, which uses recycled textiles such as denim as an insulation material. Therefore it is rapidly renewable, since cotton doesn't take long to grow new crops. Therefore it uses almost only recycled material. The main benefits are that no warning label and no safety equipment are required, it has good acoustical properties, and little energy is required for the manufacturing. It contains no toxic chemicals and comes in batts. Other advantages are that it is easy to install, non-toxic and resistant to mold, pests and fungus. Due to its non-toxic boron-based fire-retardant treatment it is also fire resistant. Also, It has good soundproofing capabilities and an R-value of 3.4-3.7 per inch. The only problem is that it is hard to find. Unlike fiberglass, there is no formaldehyde off-gassing.

Another natural, renewable and sustainable option is Wool insulation. This material can absorb moisture without compromising its thermal efficiency; when they absorb moisture they generate heat, preventing condensation in construction by maintaining temperature above dew-point in damp conditions. Additionally it is safe to touch and requires no specialized safety regulations making it easy and safe to install. Other benefits are that wool doesn't support combustion and will extinguish itself when exposed to fire and no there is no performance loss over time as wool does not settle due to elasticity.

A more sustainable option compared to traditional foam insulation is soy-based foam insulation. It is sprayed into the house in liquid form. During hardening, it expands 100 times its size, making it airtight seal as it fills all spaces during expansion. It is non-toxic and resistant to moisture, mold, fungus and rodents. Additionally it has a high R-values in less space than batt insulation. However its installation is more expensive due to its complicated spraying procedure.

V. CONCLUSION

As it stands, our current estimate for the aggregate cost of building materials is \$10,000[roofing, all materials plus installation] + \$8,644[flooring] + \$3,864[interior wall] + \$19,621 [exterior wall] = \$42,129[total] Since our total budget is \$180,000, we are within budget for costs. This does not yet include the cost of the SIPs which will be used as the main structural and insulation material, from a quick estimate, we found that the average cost to be around \$20,000 maximum. This leaves around \$120,000 leftover for building materials, we recommend the Triton Solar Decathlon Team consider more expensive materials based on their needs and if budget allows. Materials with higher price points could have better benefits than currently recommended materials.

A. Future Work

To further understand the environmental impacts, data of carbon footprints for each material could be analyzed. This data will help us analyze the environmental impacts transporting the material will yield.

In addition, a list of local manufacturers for materials that are being considered for use will be helpful in limiting carbon footprints. Another cost analysis could be done based on manufacturers for each material. This list also serves to further assess the producibility criteria for each material.

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